

RFI — Equine motion sensors with live mobile data (LTE-M / NB-IoT) (BSV EquiCare)

To: IMU / wearable-sensor vendor

From: Bharat Sports Venture — EquiCare engineering

Covers client monitoring points: steps/locomotion · general activity/abnormalities · resting pattern and duration · lameness/limb-favoring

0. How to reply — please read first

Please answer all of this in ONE consolidated reply. We are evaluating several suppliers in parallel on a fixed timeline and will not be sending rounds of follow-up questions. In practice that means:

1. **Answer every numbered question**, in the reply slots provided. If something is not supported, **write that plainly** — "not supported" is a perfectly good answer and costs you nothing. A blank or skipped item, however, we have to record as *not supported*, because we cannot tell the difference.
2. **Attach the documents and files listed in the checklist at the end.** Please send them with this reply, not "on request" — a claim we cannot verify counts as unverified.
3. **Send the actual artifact, not a description of it.** Our single worst experience with a previous supplier: they confirmed "Linux SDK — yes", and the package that arrived was x86-64 only, with no ARM64 build available at all. That cost both sides weeks. So where we ask for a binary, driver, register map or protocol spec, please attach the real file (or a download link) — or state clearly that it does not exist.
4. If a question does not apply to your product, write **"N/A"** and one line saying why.

We would rather receive an honest reply with several "not supported" answers than an optimistic one that unravels at integration.

1. What we are building

A 24/7 monitoring system for stabled horses in India. Our software turns raw motion data into steps, activity, lying time and lameness measures. **All analytics are done by us** — we are not buying your algorithms or dashboard. We need **raw, timestamped, per-axis data** and a documented way to receive it.

The data must be **live both in the barn and when the horse leaves it** — for an hour or more of exercise in an arena, on the lunge or out on a ride. So the horse's sensors must reach us over the **mobile network (LTE-M and/or NB-IoT)**, sending **directly to our own server in India**, not only to a gateway in the barn.

2. The sensor set we want quoted

We are asking for **three devices per horse**. The split keeps weight off the leg, and puts sensors where lameness is measured: horses with a sore front leg **nod the head** when that leg lands, and horses with a

sore hind leg **hike the hip** on that side. Measuring head and pelvic movement is the approach used by veterinary lameness systems.

Device	Worn	When	What it does
A. Leg tag	One front cannon bone (strap or boot pocket)	24/7	Steps, activity, standing / lying, that leg's stride timing. Light: target ≤50 g .
B. Halter hub with SIM	On the halter (cheek piece or poll strap)	24/7	Relays A and C live over LTE-M / NB-IoT . Its own IMU measures head movement (front-leg lameness). Target ≤150 g (our target — state yours).
C. Pelvis sensor	Over the sacrum (croup), on a pad, roller/surcingle or adhesive mount	Exercise and trot-ups	Pelvic movement (hind-leg lameness).

Alternative: if you offer an **all-in-one SIM leg tag**, please quote it as well, with its weight and battery life; we will compare it with the set above.

REPLY (WHICH DEVICES YOU CAN SUPPLY, AS-IS OR ADAPTED)

3. Hard requirements — please confirm these first

Pass/fail for us. If any cannot be met, please say so plainly rather than leaving it blank — it saves us both time.

#	Requirement	Why
H1	Live data over LTE-M and/or NB-IoT on Indian networks , with a fallback (e.g. LTE Cat-1 / Cat-1 bis) where those are not available. State the networks and bands you have tested in India .	The horse must stay monitored when it leaves the barn. LTE-M coverage in India is limited; NB-IoT and a fallback matter.
H2	Direct to our server: the device sends to an endpoint we configure (MQTT or HTTPS over TLS) using a documented protocol — no mandatory vendor cloud in the data path.	India DPDP-2023 residency; our data, our server.
H3	Raw per-axis time-series accessible — not only step counts or vendor-computed summaries. (Live summaries over the mobile link with raw uploaded later is acceptable — see section 6.)	Our gait / lameness models need the waveform.

#	Requirement	Why
H4	India approvals: TEC (MTCTE) where it applies to the cellular device, WPC/ETA for every radio (including the link between the devices), BIS .	Legal deployment requirement.
H5	Any SDK, driver or software for our on-site computer must be built and tested for Linux ARM64 (NVIDIA Jetson) — or use an open protocol.	A previous vendor shipped x86-only binaries and could not rebuild — it blocked us for weeks.

REPLY (H1–H5)

4. Sensing

Please answer with **specific values / part numbers**, not "yes — supported", for **each device (A, B, C)** where they differ.

1. **Sensor part number(s) + datasheet.** Is it **6-axis** (3-axis accel + 3-axis gyro) or **9-axis** (with magnetometer)? *6-axis is our minimum; 9-axis preferred.*

REPLY

2. **Accelerometer full-scale range**, and whether it is selectable ($\pm 2/4/8/16$ g). We require $\geq \pm 16$ g — a horse kicking or striking the stall wall will clip lower ranges and corrupt the very data we need.

REPLY

3. **Gyroscope full-scale range** (selectable ± 250 – 2000 °/s?). We require $\geq \pm 2000$ °/s — no clipping at trot, canter or kick.

REPLY

4. **Configurable output data rate (ODR).** We require ≥ 100 Hz; **200 Hz preferred for the head and pelvis during trot.** State supported rates, whether they can be changed **remotely**, and the achievable rate *while recording raw*.

REPLY

5. **Noise density / RMS noise** for accel and gyro at our target ODR. State any on-chip filtering and whether it can be bypassed.

REPLY

6. **Gyro bias drift with temperature** (Indian barns reach 45–50 °C), and whether a device can be **recalibrated in the field without removing it** from the horse.

REPLY

7. **On-firmware activity classification** (standing / lying / walking / trotting / restless) — do you provide it, is it **equine-validated or generic**, and **can it be disabled**? We classify ourselves, so we need raw data regardless; we do not want firmware quietly pre-filtering it.

REPLY

8. **Step counting** — equine-validated or adapted from human/cattle? State error at walk and at trot.

REPLY

9. **Per-reading quality / confidence flag** — any validity, saturation or signal-quality indicator per sample or per batch, that we can carry into our data model?

REPLY

10. **On-device summaries for the live link.** NB-IoT is too slow for continuous raw data. Can the hub (or each device) compute and send, **live**, per-minute summaries — steps, activity, lying/standing — and **per-stride** measures during trot (head and pelvis vertical-movement asymmetry, stride duration)? Or does it only forward raw data? If summaries are fixed by you, list them; if programmable, say how.

REPLY

5. The live mobile link

11. **Modem and networks.** Module part number; **LTE-M / NB-IoT / fallback** modes; **bands**; networks you have **tested in India**; behaviour when no network is available.

REPLY

12. **What is sent live, and how often.** Our target: summaries at least **once a minute** while the horse is active, and an **alert immediately** (e.g. device removed, no movement for a long time outdoors). State achievable intervals and **end-to-end latency** to our server.

REPLY

13. **Protocol to our server.** MQTT or HTTPS; **TLS** version; **configurable endpoint and credentials** (per device or per fleet); **attach the payload schema** and a sample. Confirm it can be mapped to our contract `{horseId, metric, value, unit, ts, source, confidence}`. Is the schema versioned?

REPLY

14. **SIM.** Physical SIM or eSIM; can we use **our own SIM / operator**; multi-operator profiles; who manages SIM provisioning. If you offer a connectivity platform, confirm it **does not hold or relay our data**.

REPLY

15. **Data volume and cost.** Expected **MB per horse per day** in each mode (in the barn, during exercise), and the data plan you recommend.

REPLY

16. **Link between the devices** (A and C to the hub B): radio (BLE version, Coded-PHY?), range on the horse, and what happens when the hub loses a device.

REPLY

17. **In the barn.** Is there an optional **local path** (BLE / Wi-Fi to a gateway or our on-site computer) that saves mobile data while the horse is in its stall? If so, how do the devices switch between the local path and mobile?

REPLY

6. Raw data, gaps and buffering

18. **How raw data reaches us.** The raw 100–200 Hz waveforms (especially of exercise and trot-ups): sent over the mobile link, or **offloaded in the barn** over BLE / Wi-Fi? State the **delay** before we have the raw data of a session, and the data volume per hour of raw.

REPLY

19. **On-device buffering** — we require **≥ 24 h on every device** with **automatic backfill using the original timestamps** and acknowledged delivery. State capacity in **hours at 100 Hz raw**, and whether it stores **raw or only summaries** when it cannot send.

REPLY

20. **The exercise scenario.** A horse leaves the barn for **1–2 hours** of exercise; mobile coverage in the arena or on a ride is **patchy**. State exactly: what is **sent live**, what is **stored**, how and when **gaps are filled**, and whether anything can be **lost**.

REPLY

7. Time synchronisation

21. **Timestamps on the device** or on receipt? State the **time source** (network time, NTP over the mobile link, GNSS), worst-case **drift over 24 h** while out of coverage, and how closely the **three devices on one horse are aligned** with each other. *We relate head and pelvis movement to the leg's stride, so we are targeting ≤ 10 ms between devices, and tens of milliseconds against our camera video.*

REPLY

8. Location (optional)

22. **GNSS in the hub** for position, speed and distance during exercise and rides: available? Accuracy, power cost, and can it be **switched off** (privacy)?

REPLY

9. Power

23. **Battery life for each device in each mode** — hub with live reporting every 1 minute and every 5 minutes; power-saving (PSM / eDRX) support; leg tag and pelvis sensor at 100–200 Hz raw recording.

Chemistry/mAh; rechargeable or replaceable; charge time and cycle life.

REPLY

24. **Charging operations for a whole yard** — multi-bay dock, hot-swap spares, how data gaps during a swap are handled, and how a replacement device is **re-bound to the same horse**.

REPLY

10. Fit, welfare and environment

25. **IP rating** — **we require IP67 or better** on all three devices, validated against wash-down, mud, urine/manure and sweat. State operating temperature and humidity range (we need 45 °C+ and monsoon).

REPLY

26. **Attachment** — leg tag to the cannon bone (sizes **pony** → **draft**), hub to a standard halter, pelvis sensor over the sacrum (pad, roller or adhesive). Expected strap and mount service life.

REPLY

27. **Welfare** — weight and dimensions of each device, and any 24/7 wear assessment showing no rub sores or skin damage (leg and halter). Please share equine field-trial or welfare data.

REPLY

28. **Loss prevention** — **tamper / detach alert**, sent **live by the hub** when a leg tag or pelvis sensor goes silent or comes off; your observed **device-loss rate per device-year**.

REPLY

11. Identity and fleet management

29. **Unique immutable hardware ID** in every record (plus IMEI/ICCID for the hub); how the three devices are **paired** and how the set is **bound to a horseld** via API.

REPLY

30. **Fleet management** — firmware **OTA over the mobile link**, battery/health monitoring, remote configuration (sample rates, reporting interval), and what data (if any) passes through systems you host.

REPLY

12. Evaluation

We would like **3–5 leg tags + 2 halter hubs (with SIMs or eSIM profiles that work in India) + 1–2 pelvis sensors + a charging dock**, together with the **protocol documentation, payload schema and sample code**, so we can validate on real horses before committing to a design.

REPLY (AVAILABILITY + LEAD TIME FOR EVALUATION HARDWARE)

13. Optional — equine validation data

If you hold any **equine** reference or validation data — step counts, lying time, gait asymmetry, or **lameness measures against a veterinarian's lameness grading** — please share it. Most devices are validated on cattle or humans; equine gait differs, and this would materially reduce our modelling risk.

REPLY

Response checklist

Please confirm each item is attached or answered, so we can evaluate in a single pass:

	Item
☐	Every numbered question answered (or marked "not supported" / "N/A")
☐	All hard requirements (H-numbers) explicitly confirmed or declined
☐	Sensor / component datasheets for each device, and the modem module datasheet
☐	Protocol document and payload schema for the live link — the actual file
☐	Sample live payload and sample raw data, showing real field names and units
☐	Networks and bands tested in India , and TEC / WPC / BIS certificates (or their status)
☐	Expected mobile data per horse per day
☐	ARM64 (aarch64) build of any on-site software, or a plain statement that none is needed / none exists
☐	Sample code, if you provide any
☐	Evaluation-unit availability and lead time
☐	A named technical contact we can reach directly with any single clarification

If anything on this list genuinely cannot be shared before an order, please say which item and why, so we can factor it in rather than assume the worst.

Commercial terms are not part of this RFI — we will discuss those separately once the technical fit is confirmed.